

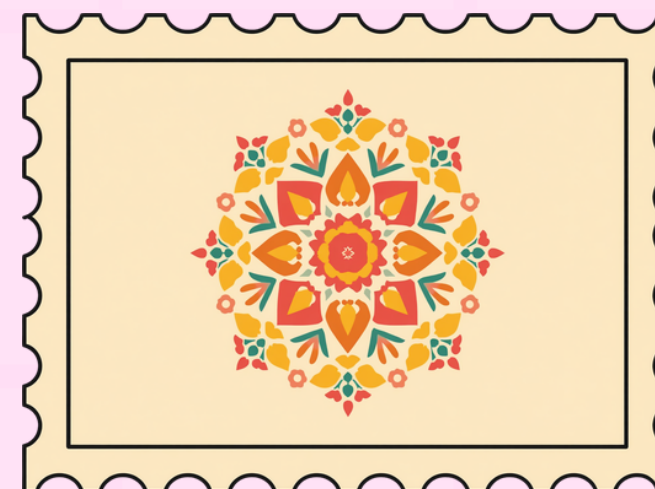
PYTHON - TURTLE ART: GEOMETRIC PATTERNS





Today's Digital Rangoli!

Today YOU will make the computer draw geometric art – like a digital rangoli – using Python! Let's have fun creating art with code!





What is Python?

Python is a language that computers understand – like mummy's recipe card that tells exactly how to cook your favourite dish! We will tell the computer what to do, step-by-step!





What is Turtle?

Turtle is a tiny robot tortoise that holds a pen! Just like drawing rangoli or kolam patterns on the floor, we will use Turtle to draw beautiful shapes on the computer screen.



Core Commands to Know



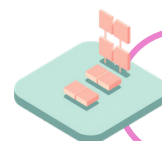
`import turtle`
Bring the turtle toolbox



`turtle.right(90)`
Turn right by 90 degrees



`turtle.penup()/pendown()`
Lift or place the pen



`turtle.forward(100)`
Walk 100 steps forward

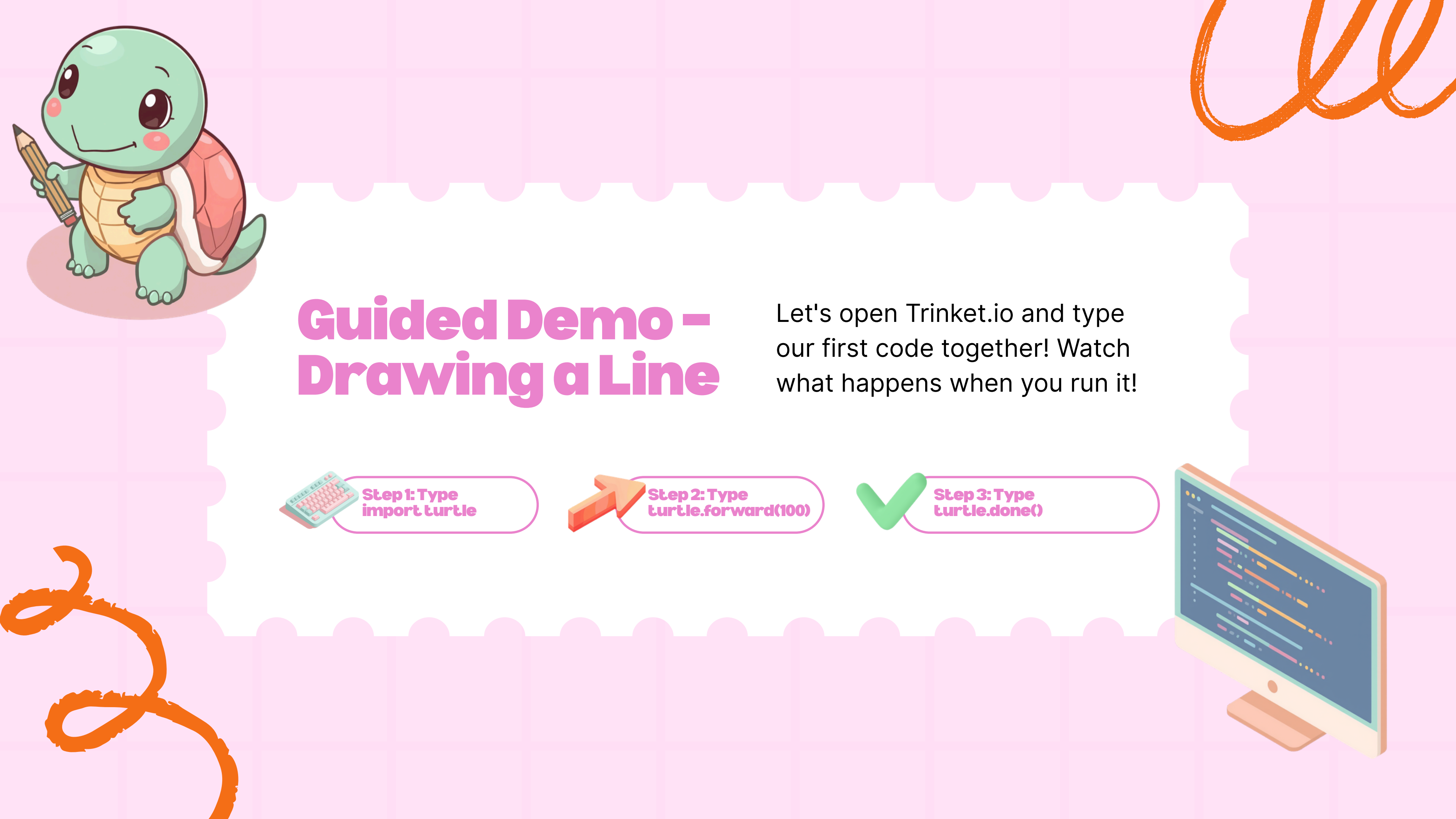


`turtle.left(90)`
Turn left by 90 degrees



`turtle.done()`
Freeze the drawing!





Guided Demo - Drawing a Line

Let's open Trinket.io and type
our first code together! Watch
what happens when you run it!



Step 1: Type
`import turtle`



Step 2: Type
`turtle.forward(100)`



Step 3: Type
`turtle.done()`

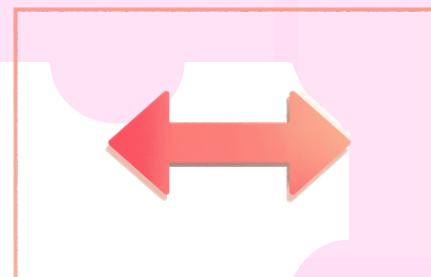




Run Your First Code!

100 means 100 steps – the turtle walks forward that much! Challenge: What if we try `forward(200)` instead? Try and see what happens!





Guided Demo – Drawing a Square (Step 1 & 2)

A square has 4 sides. After each side, turn right 90 degrees. Let's make the turtle walk and turn like you do when you play Ludo!

```
turtle.forward(100)
```

```
turtle.right(90)
```

```
turtle.forward(100)
```

```
turtle.right(90)
```





Drawing a Square – Complete Code!

Four sides, four turns – a perfect square! Type this code and watch your turtle draw a beautiful square. Brilliant! You're coding a shape now!





Physical Activity - Walk the Square!

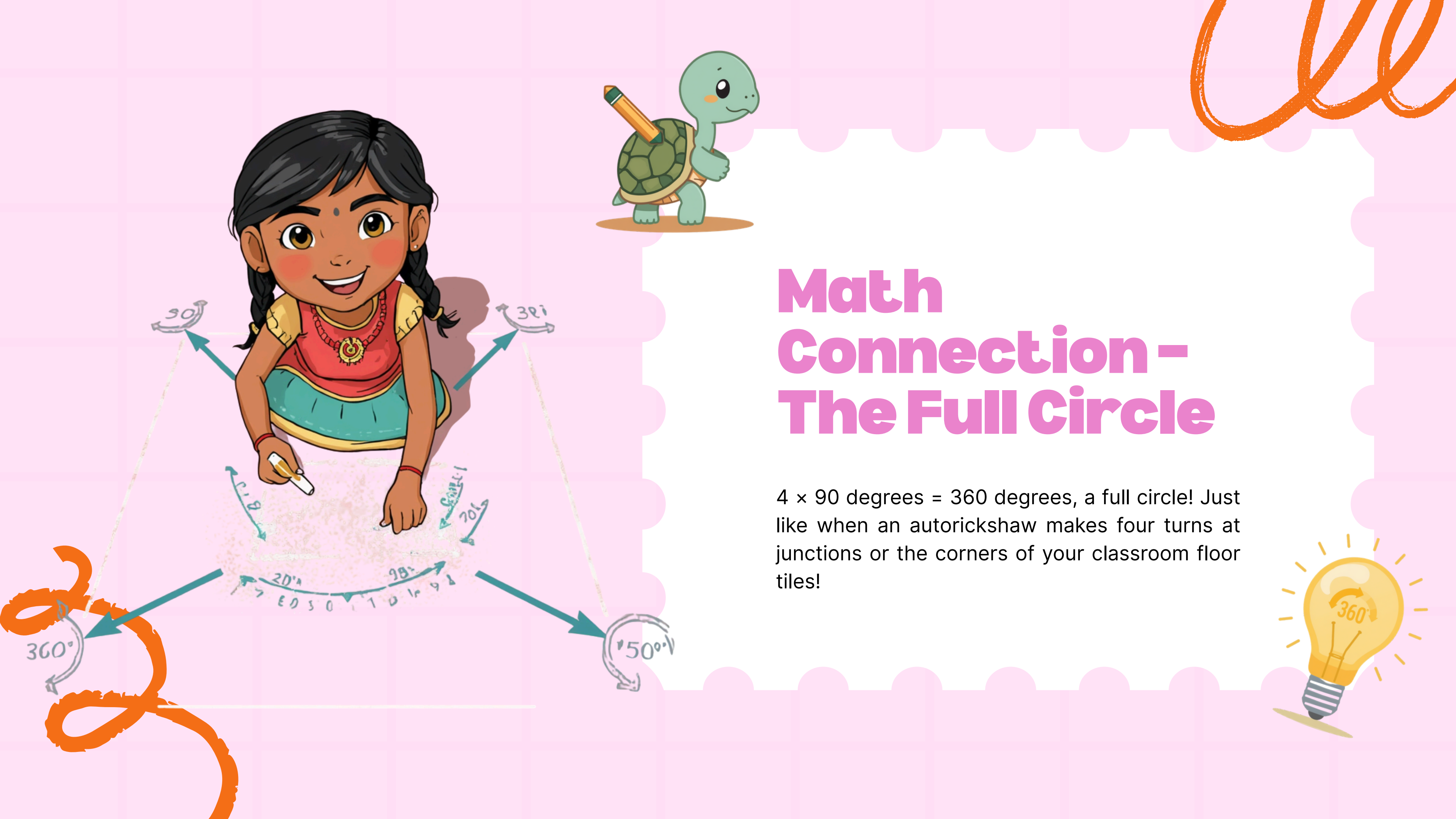
Stand up! Walk 3 steps forward. Turn RIGHT 90 degrees. Repeat 4 times to make a square! This is exactly what your turtle does in code. Let's move like a turtle!





Math Connection – The Full Circle

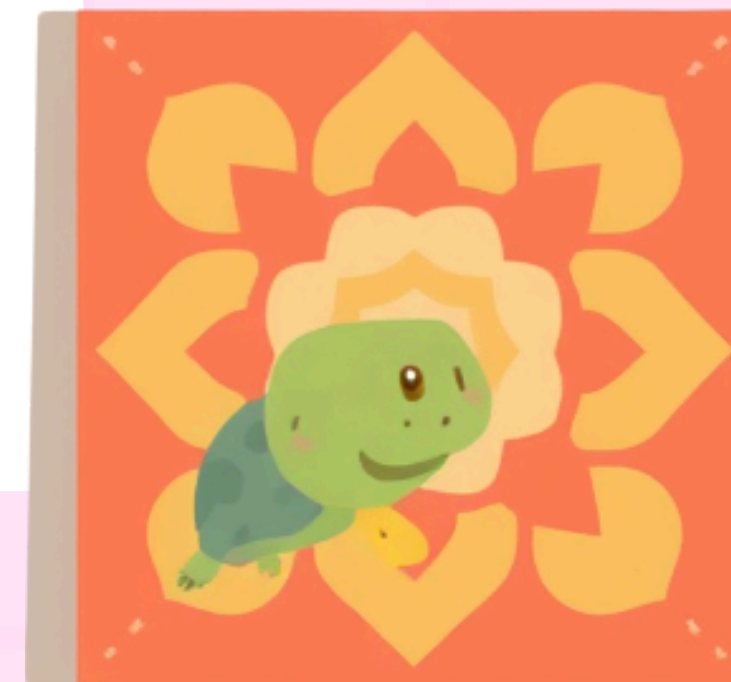
$4 \times 90 \text{ degrees} = 360 \text{ degrees}$, a full circle! Just like when an autorickshaw makes four turns at junctions or the corners of your classroom floor tiles!



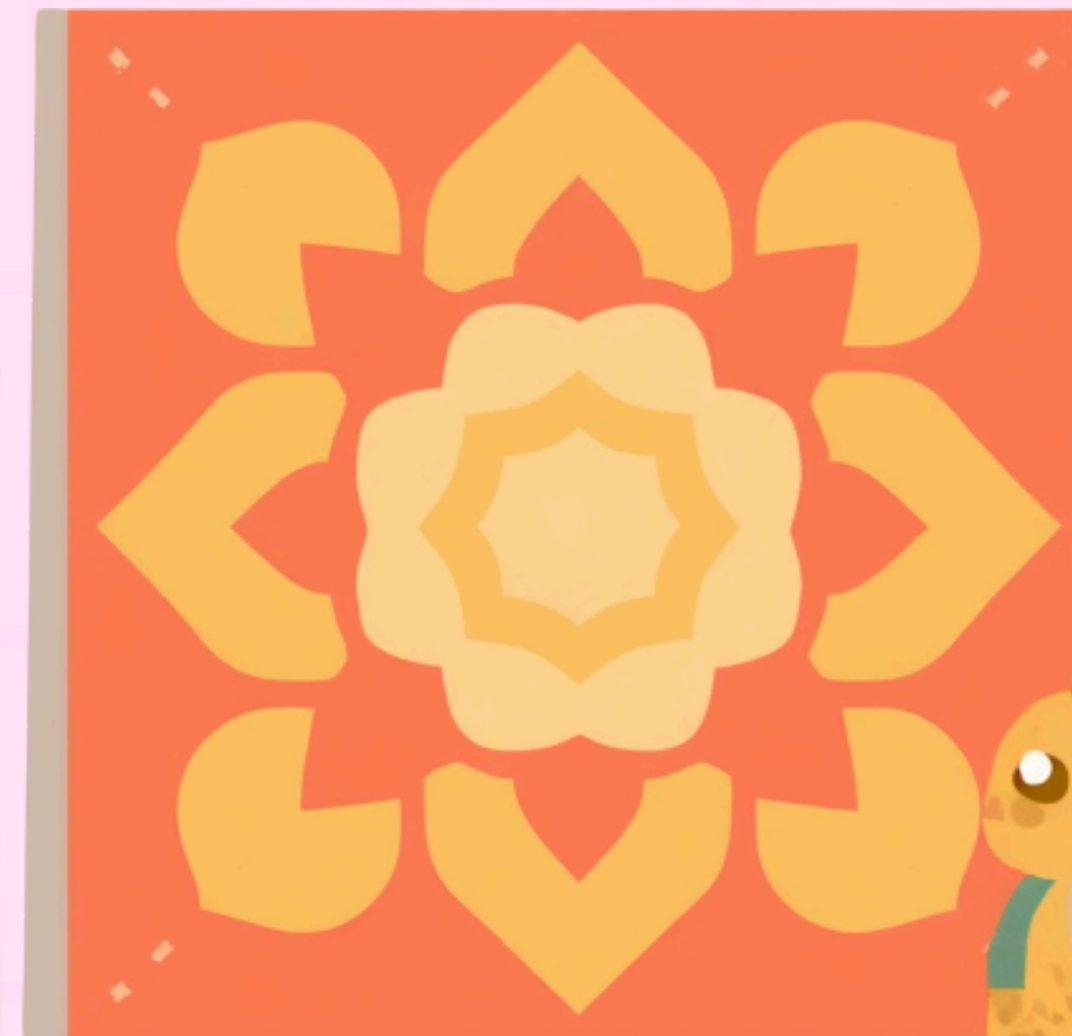


Challenge - Change Square Size!

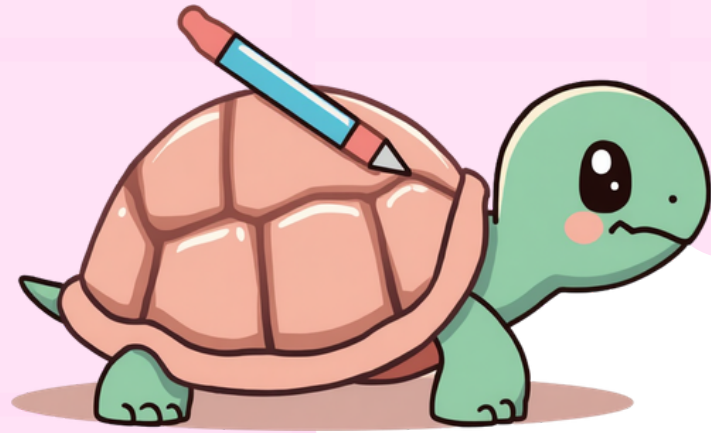
What if we change `forward(100)` to `forward(150)`? Try it yourself and see what happens! You're learning to control the turtle's steps!



100 steps

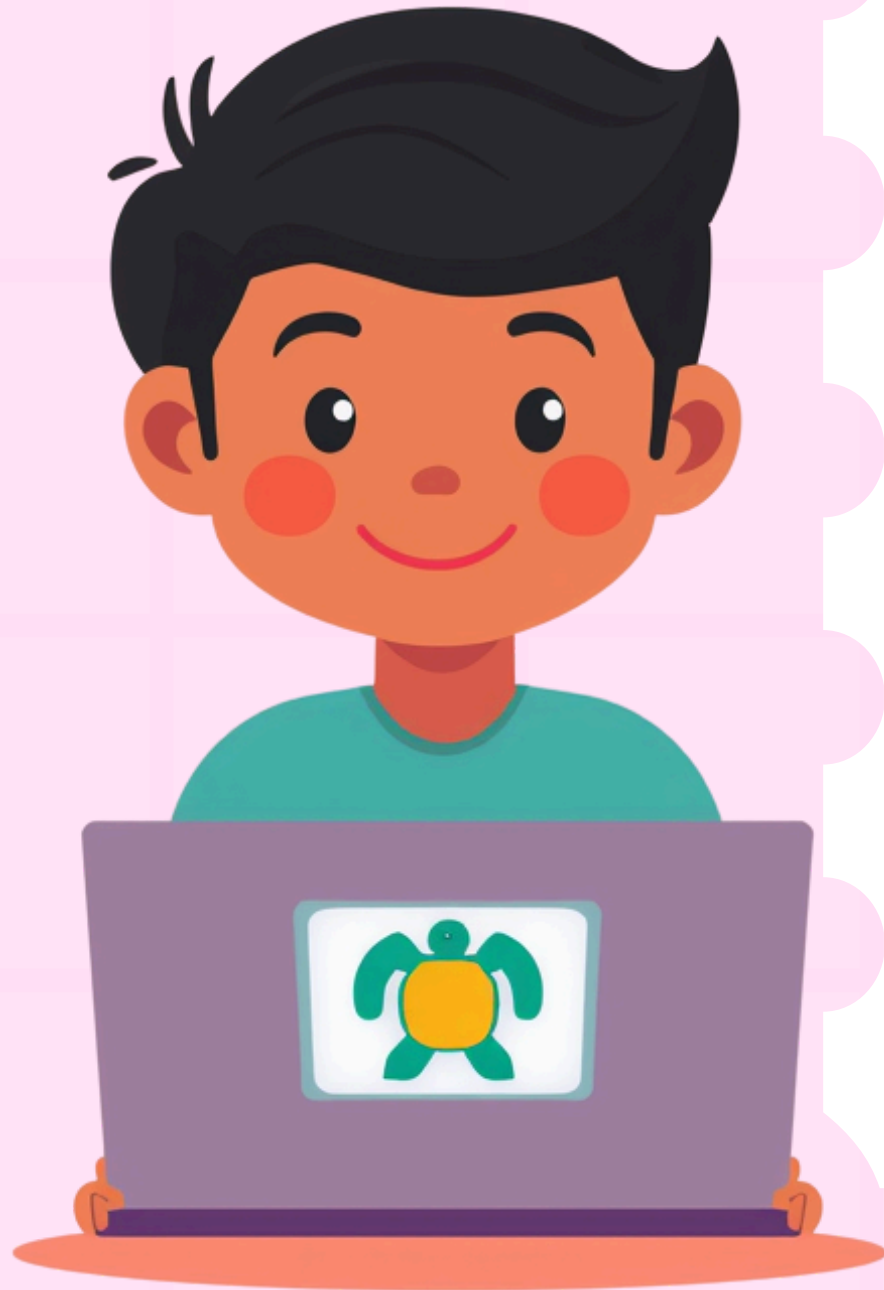


150 steps



Student Activity Time – Draw YOUR Square!

Now it's your turn! Draw your own square using the turtle. Use forward and right commands. You're doing so well, keep coding!

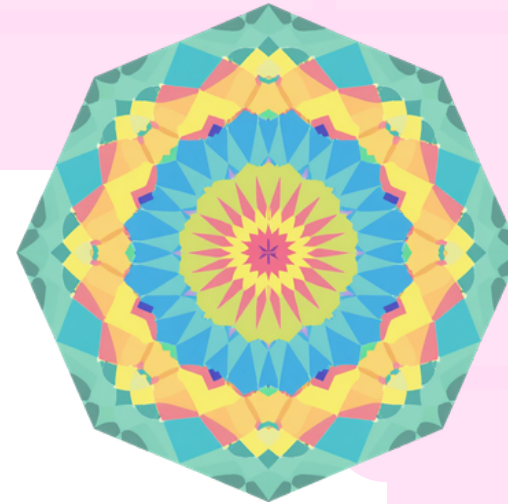




Early Finishers - Two Squares!

Try making TWO squares side-by-side! Use `turtle.penup()` and `turtle.pendown()` to move without drawing. Amazing creativity, Arjun and Meera!





Bonus Challenge – Draw a Hexagon!

Try `turtle.right(60)` instead of `turtle.right(90)`.
What new shape did you make? Wow! You're
discovering new shapes like a real artist!





Review Quiz - Questions 1 & 2

Let's see what you learned today!
Think carefully and answer these
fun questions about turtle
commands!



**Q1: Which command
makes turtle move
forward?**

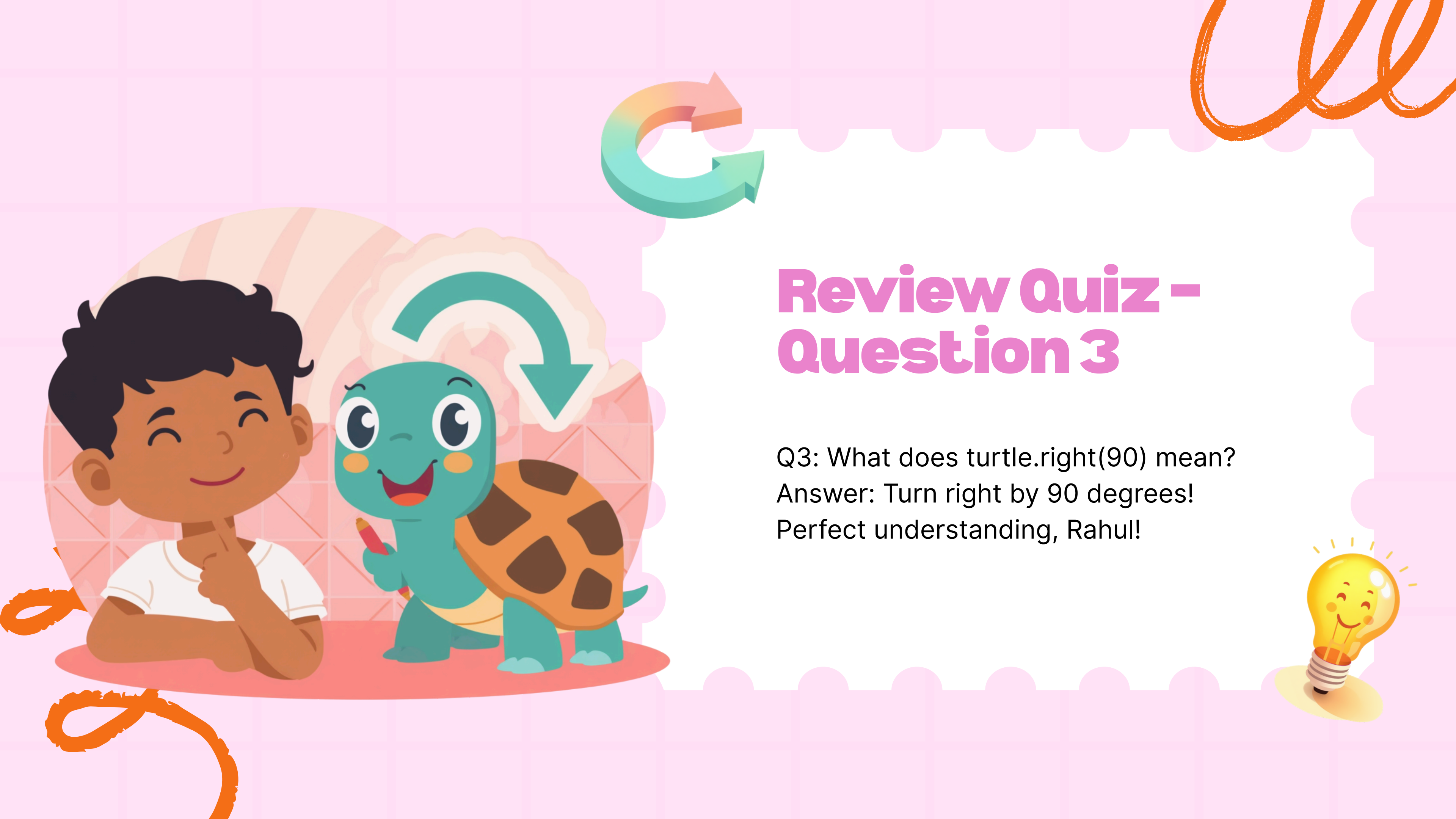


**Answer:
turtle.forward()**



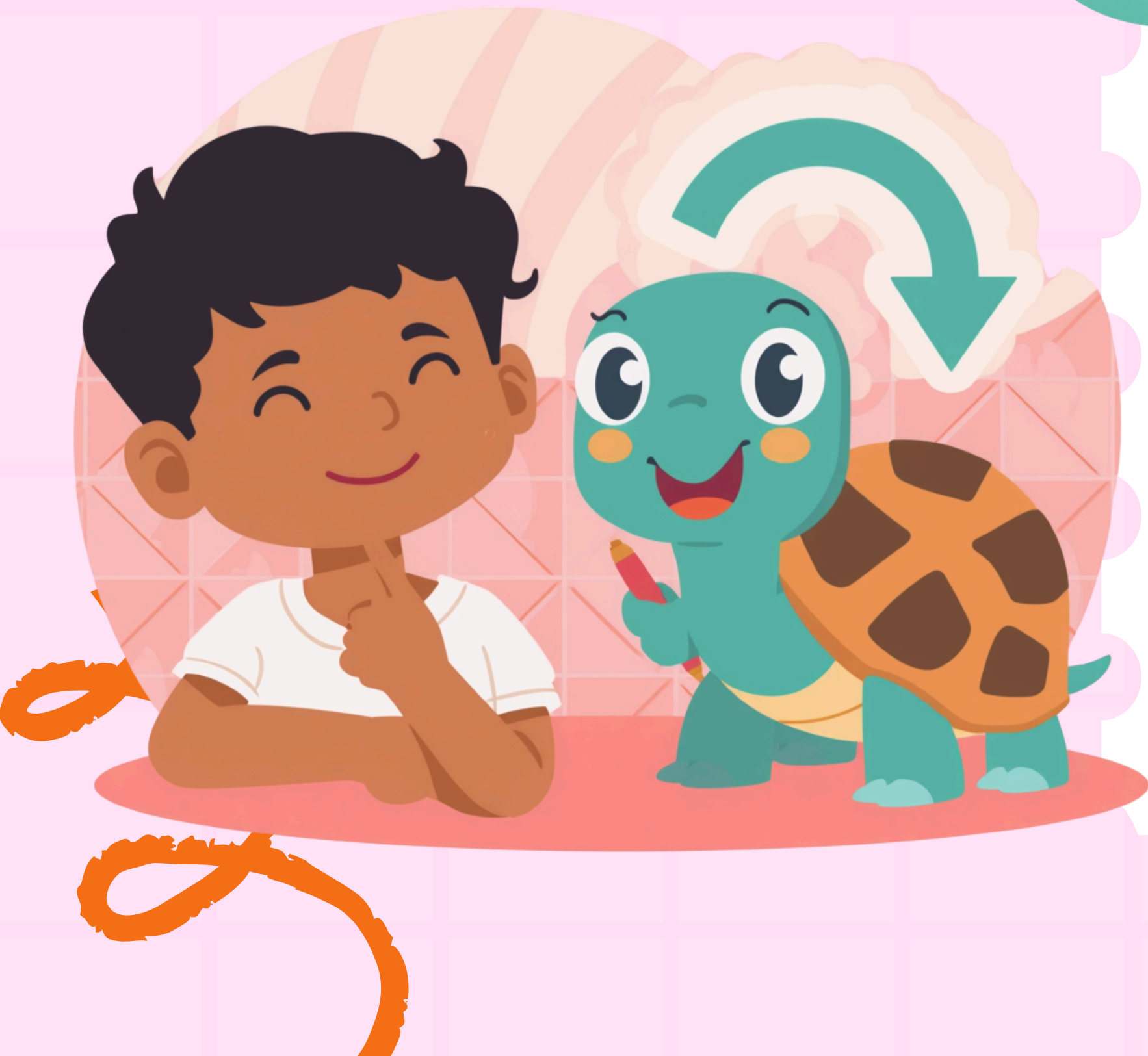
**Q2: How many right turns for
a square? Answer: 4 times!**





Review Quiz - Question 3

Q3: What does `turtle.right(90)` mean?
Answer: Turn right by 90 degrees!
Perfect understanding, Rahul!



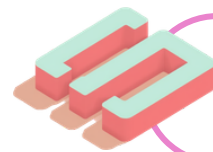


Wrap-Up & Homework

Great job today! You learned Python Turtle basics, drew lines and squares, and used forward and right commands. Now practice at home!



Draw a square on paper



Write `forward(100)` and `right(90)`



Next: Add COLOUR!





Indian Cultural Inspiration

Just like traditional rangoli art, your turtle art is geometric and beautiful! Proudly inspired by Indian heritage!





**THANK YOU &
KEEP CODING!**

